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Background and Motivation

Optical metasurfaces are bidimensional arrays of metallic/dielectric nanostructures —known as meta-atoms— specifically tailored to behave in a way no found in nature when illuminated at specific wavelengths [1, 2]. Depending on the physical properties of the meta-atoms, that is, their composition, size, shape, orientation and distribution within the bidimensional array [1, 3], metasurfaces allow to shape at will the spatial optical response of the system [4], thus suiting them for a variety of applications in fields such as spectroscopy [1], color structuration [2], communications [4], and sensing [1, 3–6]. In the last decades, the interest in optical metasurfaces for medical applications has increased due to the need for sensitive, fast, low-cost and easy-to-use technologies [3, 5], like metasurfaces with plasmonic (metallic) meta-atoms used as contrasting agents for bioimaging [3], and as free-label biosensors returning real-time measurements [1, 5, 7].

Metasurfaces designed for biosensing typically consist of a nanostructurated substrate with compatible microfluidic devices illuminated with a white light source and a light recollection system, allowing for scattering or extinction measurements, followed by an spectrometer [5, 8]. One particular kind of biosensing-aimed metasurfaces consists in plasmonic meta-atoms, exploiting their property of high confinement of light at nanometric scales, yielding an improvement in the sensitivity of various detection techniques [1]. The light confinement is the result of the meta-atom's Localized Surface Plasmon Resonances (LSPRs) being excited at the meta-atom's interface with its surroundings, which occurs when the electromagnetic fields couple to the free electrons of the plasmonic structure [3–5]. Since the LSPR is material and geometry dependent, a variety of plasmonic metasurfaces have been designed [7–10] —each with its own benefits and disadvantages [4, 5]— as those shown in Fig. 1, all of which are plasmonic metasurfaces consisting of gold (Au) meta-atoms on a glass substrate but with different geometries and distributions within the metasurface. For example, Feuz et al. [8] employed a short-range ordered metasurface of nanoholes to sense protein binding events in real-time [Fig. 1a)], while Kabashin et al. [7] measured changes in the refractive index of the media embedding an ordered metasurface of plasmonic nanorods [Fig. 1b)]. Metasurfaces with simpler geometries and distributions can be used for biosensing as well, as shown by Qiu et al. [9], who employed a disordered metasurface of nanospheres to detect selected DNA sequences from Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) [Fig. 1c)], or by Svedendahl et al. [10], who sensed protein binding events with a short-range ordered metasurface of nanospheres [Fig. 1d)].

Since LSPRs are excited at frequencies that depend on the material and geometry of the nanostructures, metasurfaces of various configurations have been designed [4, 5, 7–10], as shown in Fig. **??**. For example, Kabashin et al. [7] reported a biosensing metasurface consisting of

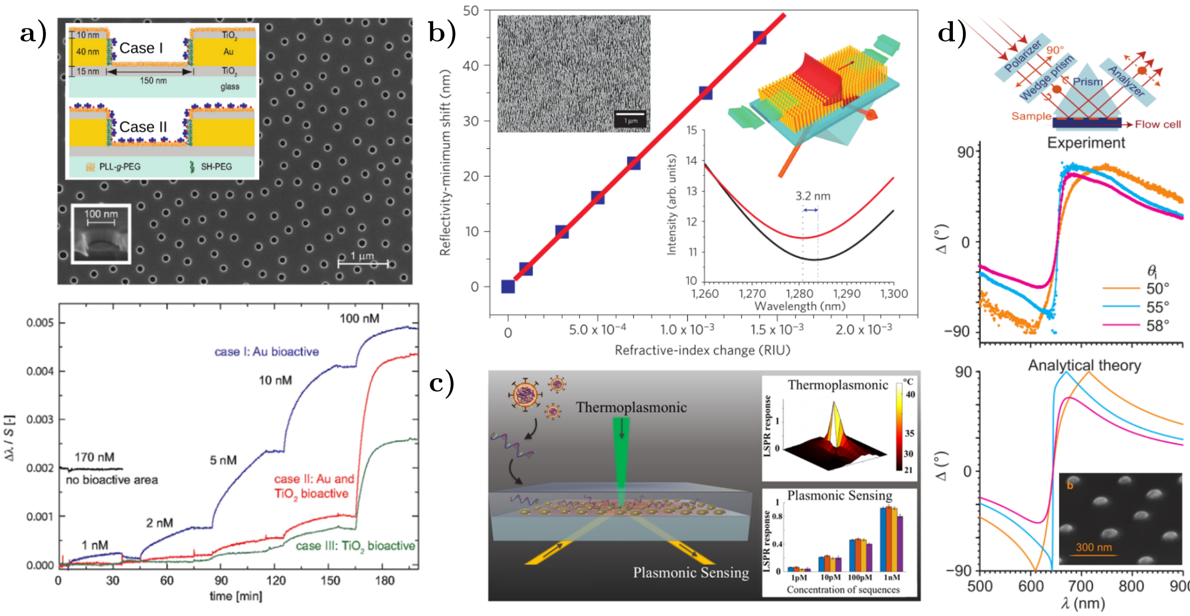

Fig. 1: Examples of biosensing-aimed plasmonic metasurfaces. **a)** Short-range ordered metasurface of nanoholes in a Au film [Scanning Electron Microscopy (SEM) image and meta-atom scheme] and real-time measurements of the LSPR redshift due to protein binding events; images extracted and adapted from [8]. **b)** Reflectivity minimum shift of an ordered metasurface of Au nanorods (SEM image and scheme in the inset) as a function of the refractive index change of the media embedding the metasurface; images extracted and adapted from [7]. **c)** Schematics of a disordered metasurface of Au nanospheres designed for SARS-CoV-2 detection and the LSPR response of its meta-atom: Thermoplasmonic and plasmonic sensing; image extracted from [9]. **d)** Experimental and theoretical results for the ellipsometric parameter Δ as a function of the incident wavelength, when a short-ranged ordered metasurface of Au nanospheres (SEM image) is illuminated by a non-polarized white light as shown in the setup diagram; extracted and adapted from [10].

cylindrical gold nanoparticles (AuNPs) arranged in a square lattice [see Fig. **??**]; this metasurface exhibited appropriate sensitivities for detecting changes in the refractive index of aqueous samples. Another example of the versatility and advantages of plasmonic metasurfaces for biosensing is a disordered array of spherical AuNPs on a glass substrate, used by Svedendahl et al. [10] [see Fig. **??**] and Qiu et al. [9] [see Fig. **??**], who detected real-time protein binding events and SARS-CoV-2 DNA sequences, respectively, by illuminating the metasurface in an internal incidence configuration and measuring the ellipsometric parameters [9, 10]. It is worth noting that with a random distribution of NPs, fabrication processes such as laser ablation [11] or thermal treatment known as *dewetting* [**cuanalo_sensitivity_2022**, 12] can be employed, reducing fabrication time and cost.

Metasurface design can be achieved using both analytical and numerical methods to estimate the physical characteristics that optimize their optical response for a given application. For example, the resonance wavelengths of LSPRs in ordered metasurfaces of arbitrary nanostructures have been calculated using the Finite Element Method (FEM) [8] or the Finite-Difference Time-Domain (FDTD) algorithm [13]. Additionally, the optical response of disordered metasurfaces of spherical NPs has been analytically studied using effective medium theories under the small-particle approximation [14]. These models describe two-dimensional arrays as equivalent continuous media [15].

The motivation for such a physical system arises from the collaboration between two experimental research groups[1] and one theoretical research group[2]. The experimental groups have fabricated disordered arrays of partially embeded Au nanospheres with an average radius of 12.5 nm —a potential meta-atom for biosensing-aimed metasurfaces—.

During the design stage of metasurfaces, both analytical and numerical methods can be employed to estimate the physical characteristics that optimize their optical response based on the intended application. For instance, the wavelengths at which LSPRs of ordered metasurfaces composed of arbitrary nanostructures are excited have been calculated using the Finite Element Method (FEM) [8] or the Finite-Difference Time-Domain (FDTD) algorithm [13]. Likewise, the optical response of disordered metasurfaces consisting of spherical NPs has been analytically studied using the effective medium theories under the small-particle approximation [14], where the response of two-dimensional arrays is modeled as an equivalent continuous medium [15]. Examples of effective medium theories include the Dipolar Model (DM) [14, 16], the Island Film Theory (IFT) [10, 17, 18], and modifications to the Maxwell Garnett Model [7].

The characterization of a metasurface involves determining its physical properties, such as geometry, material composition, and spatial distribution of its nanostructures. One way to perform characterization is through imaging techniques such as Transmission Electron Microscopy (TEM), Scanning Electron Microscopy (SEM), and Atomic Force Microscopy (AFM). However, these methods often lead to sample damage or partial destruction. An alternative to these invasive methods is to compare optical measurements with theoretical results—obtained through numerical or analytical calculations—so that the calculated response matches the experimental data for specific metasurface parameters. For example, since the spectral position of the LSPR depends on the geometry and size of an NP, it can be identified by comparing reflectance measurements with estimated values. It is important to note that this approach requires the theoretical conditions to be reproducible in a laboratory setting.

In the case of disordered metasurfaces, effective medium theories such as DM or IFT are suitable for describing the optical response of metasurfaces composed of spherical NPs—small relative to the wavelength of illumination—perfectly positioned on a dielectric substrate [16, 17]. Both DM and IFT describe each spherical NP in the two-dimensional array as an electric dipole, allowing for relaxation of the perfect spherical geometry assumption by introducing depolarization factors, thereby extending these models to ellipsoids [19]. Additionally, DM and IFT can be applied without modification even if partial embedding of the NPs into the substrate is identified, as long as it does not exceed 12% of the NP's volume[3]. Although certain geometric and embedding conditions can be relaxed while still accurately describing the behavior of disordered metasurfaces, this does not apply to all assumptions of DM or IFT, such as modeling all NPs as electric dipoles.

[1]The Biophotonics Group and the Organic and Hybrid Semiconductor Optoelectronics Group at *Instituto Nacional de Astronomía, Óptica y Electrónica* (INAOE).

[2]The Nanoplasmonics Group at *Facultad de Ciencias, Universidad Nacional Autónoma de México* (UNAM).

[3]In my master's thesis, FEM calculations of absorption and extinction cross-sections of a 12.5 nm radius AuNP illuminated in the visible spectrum showed that the LSPR shift relative to the case of a perfectly positioned NP was at most 5 nm when the embedding depth was one-eighth of its volume.

Both DM and IFT require the small-particle approximation, meaning a disordered metasurface of spherical NPs must have a narrow size distribution. However, if NP synthesis is performed using physical methods such as laser ablation or *dewetting*, the size distribution may exhibit a normal distribution with significant standard deviation [11] or a log-normal distribution [12, 20], as illustrated in Fig. **??**. In such cases, some NPs within the metasurface may be large enough to require higher-order multipolar contributions beyond the dipolar approximation to accurately describe their optical response. When the small-particle approximation does not hold for a real sample, scattering theories can be used, which do not require the small-particle approximation and, unlike effective medium theories, do not describe the metasurface as an equivalent continuous film but rather provide an approximate solution to Maxwell's equations [15, 21]. A scattering theory that describes disordered arrays of spherical NPs with a given size distribution is the Coherent Scattering Model, which can be used to design and characterize metasurfaces of spherical NPs under more realistic experimental conditions. The following section presents the hypotheses, characteristics, and expressions of this model. The proposed project for the doctoral program at PCF arises from a theoretical-experimental collaboration with the Biophotonics Group—led by Prof. Rubén Ramos García—at the National Institute of Optics and Electronics (INAOE). This collaboration motivated my research topic during the PCF master's program and now aims to extend it by introducing increasingly realistic conditions in the theoretical description of metasurfaces. The general objective of the doctoral project is to develop semi-analytical extensions to the theories mentioned in Section **??**—to describe metasurfaces in a realistic biosensing environment—and to compare them with experimental results, given that the Biophotonics Group has techniques for synthesizing disordered metasurfaces of spherical gold NPs and an experimental setup for real-time reflectance spectrum measurements [**cuanalo_sensitivity_2022**].

The goals for the first year of the doctoral program are as follows:

- **Numerical implementation of the CSM considering size distribution**
  Implement the CSM expressions given by Eq. (1.12) for a size distribution determined by micrograph images. This code should account for the size correction of the dielectric function [22, 23] of the spherical NP material to correctly compute the average size distribution.

- **Implementation of a parameter fitting method**
  Implement the gradient descent method [24] to minimize the error between optical response calculations and experimental measurements. This tool should be applicable to any theoretical model considered.

- **Comparison and evaluation of measurement schemes with the CSM**
  Evaluate and compare the method for quantifying the optical response of the metasurface when changing the refractive index of the material in contact with it. This evaluation will be conducted by comparing the figure of merit [5] for the following methods: minimum in the reflectance spectrum and phase shifts of ellipsometric parameters [9, 10].

The first goal will enable the modeling of metasurfaces like those shown in Fig. **??**, allowing the determination of optimal parameters for metasurface-based sensing applications, as well as setting a lower bound for the accuracy of the employed synthesis methods. The second goal will quantify the extent to which a model or theory describes the optical response of a metasurface while providing a tool for non-destructive metasurface characterization. The final goal focuses on analyzing the sensitivity of the metasurface and the feasibility of performing the corresponding measurements by quantifying the figure of merit of the two proposed schemes. These goals not only implement general methodologies and tools to evaluate the methods and approaches to be applied later, but also establish a path for a first publication on the characterization and performance of this type of metasurface in sensing, encompassing a theoretical-experimental analysis from synthesis to application.

Chapter 1

Theoretical frameworks and techniques

The study of wave propagation in complex media has long been a central topic in optics, acoustics, and condensed matter physics. In particular, understanding the optical response of ensembles of nanoparticles (NPs) has garnered significant attention due to its relevance in fields such as nanophotonics, plasmonics, and metamaterials. The Multiple Scattering Theories (MSTs) are consists of theoretical frameworks employed to model these systems. The MSTs explicitly accounts for the interactions between individual scatterers and provides an accurate description of wave propagation in structured and disordered media [16, 21, 25]. The Foldy-Lax equations, formulated in the mid-20th century, offer a rigorous way to model the self-consistent field scattered by a collection of particles. Another approach to solve the Maxwell's equation approximately is the formalism of excess current and charge densities, and fields —most commonly found in the context of thermodynamics—, which is suited specifically when the particles are located in an ambient and supported on a substrate.

In the following sections, two MSTs are presented in addition to experimental setups to determine the optical response of ensembles of NPs supported on a substrate. In Section 4 the Coherent Scattering Model (CSM), based on the Foldy-Lax approach, is derived, while in Section 1.2 the Thin Film Theory is presented, which is founded on the excess quantities formalism. Both MTSs are suited to model the optical response of metasurfaces conformed by ensembles of spherical nanoparticles, whose centers are randomly located but contained within a a plane parallel to a substrate supporting each element of the ensemble. Lastly, in Section 3.2 a setup for extinction spectra and one for reflectivity spectra measurements are described.

1.1 Coherent Scattering Model

Multiple Scattering Theories (MSTs) provide approximate solutions to Maxwell's equations when considering the problem of light scattered by an ensemble of particles illuminated by a monochromatic plane wave [16, 21, 25]. In particular, the Coherent Scattering Model (CSM) is a MST that provides an expression for the amplitude coefficients of reflection and transmission of light scattered in the coherent direction for a disordered ensemble of spherical particles, whose centers are coplanar [21, 26, 27].

Within the MST framework, the optical response of a disordered array of arbitrary particles, illuminated by an incident monochromatic plane wave $\mathbf{E}^{\text{inc}}(\mathbf{r},\omega)$ with angular frequency ω and propagation direction $\hat{\mathbf{k}}^{\text{inc}}$, is determined by the excitation field $\mathbf{E}^{\text{exc}}(\mathbf{r},\omega)$ acting on the k-th particle in a collection of N particles. This field is equal to the sum of the incident electric field and the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r},\omega)$ —which includes both the field inside the particle and the scattered field outside of it— due to the other elements of the ensemble but not itself [26, 27], that is

$$\mathbf{E}_k^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell\neq k}^{N} \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}), \tag{1.1}$$

where the harmonic dependence is intrinsically implied. Since all particles are excited by the same interaction, the electric field induced in the ℓ-th particle is given by [26, 27]

$$\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) = \int \mathrm{d}^3r' \mathbb{G}(\mathbf{r},\mathbf{r}') \int \mathrm{d}^3r'' \mathbb{T}(\mathbf{r}'-\mathbf{r}_\ell, \mathbf{r}''-\mathbf{r}_\ell)\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \tag{1.2}$$

where $\mathbf{r}_\ell$ is the position of the center of the ℓ-th particle, $\mathbb{G}(\mathbf{r},\mathbf{r}')$ is the dyadic Green function —solution to the vectorial Helmholtz equation with the product of the unitary dyadic and a Dirac delta centered at $\mathbf{r}'$ as a source [28]— and $\mathbb{T}$ is the transition operator, or matrix T, that linearly relates the scattered electric field of a particle to the electric field that excites it [28]. The integrals in Eq. (1.2) are performed over a volume V, described by the positions $\mathbf{r}'$ and $\mathbf{r}''$, such that the center $\mathbf{r}_\ell$ of the ℓ-th particle is located within V [26].

The total electric field $\mathbf{E}(\mathbf{r})$ is equal to the sum of $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and the N contributions of the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})$ for each particle, which is determined by solving the system of N equations given by Ecs. (1.1) and (1.2). For an ensemble of randomly located particles, the configurational average of the total electric field $\langle\mathbf{E}(\mathbf{r})\rangle$ is calculated by considering the probability of occurrence of each combination in which the centers $\mathbf{r}_\ell$ of the elements can be located [26, 27]:

$$\langle\mathbf{E}(\mathbf{r})\rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^{N}\langle\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^{N}\left(\prod_{k=1}^{N}\int \mathrm{d}^3r_k \rho(\mathbf{R})\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\right), \tag{1.3}$$

with $\rho(\mathbf{R})$ is the probability density that the ensemble is in a specific spatial configuration given by $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)^T$. Configurationally averaging Eq. (1.2) gives the average contribution of the induced electric field by the ℓ-th particle, which can be written as [26, 27]

$$\langle\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\rangle = \int \mathrm{d}^3r' \mathbb{G}(\mathbf{r},\mathbf{r}') \int \mathrm{d}^3r'' \int \mathrm{d}^3r_\ell \rho(\mathbf{r}_\ell)\mathbb{T}(\mathbf{r}'-\mathbf{r}_\ell, \mathbf{r}''-\mathbf{r}_\ell)\langle\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'',\mathbf{R})\rangle_\ell, \tag{1.4a}$$

$$\langle\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'',\mathbf{R})\rangle_\ell = \prod_{\substack{k=1\\k\neq\ell}}^{N}\int \mathrm{d}^3r_k \rho(\mathbf{R}|\mathbf{r}_\ell)\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \tag{1.4b}$$

where $\rho(\mathbf{r}_\ell)$ is the probability density of finding the center of the ℓ-th particle within the volume V, and Eq. (1.4b) represents the configurational average of the excitation electric field $\mathbf{E}_\ell^{\text{exc}}$ for the ℓ-th particle, considering that its position is fixed in V; the conditional probability density $\rho(\mathbf{R}|\mathbf{r}_\ell)$ describes such constriction. This expression can be further analyzed by substituting Eq. (1.1) into Eq. (1.4b) and following an analogous procedure to that of Eqs. (1.4) with the index exchanges $k \to \ell$ and $\ell \to m$, yielding

$$\begin{aligned}\langle \mathbf{E}^{\mathrm{exc}}_{\ell}(\mathbf{r}'', \mathbf{R})\rangle_{\ell} =& \mathbf{E}^{\mathrm{inc}}(\mathbf{r}'') + \sum_{\substack{m=1\\m\neq\ell}}^{N} \int \mathrm{d}^3 r' \mathbb{G}(\mathbf{r}', \mathbf{r}'')\times \\ & \int \mathrm{d}^3 r''' \int \mathrm{d}^3 r_m \rho(\mathbf{r}_m) \mathbb{T}(\mathbf{r}' - \mathbf{r}_m, \mathbf{r}''' - \mathbf{r}_m) \langle \mathbf{E}^{\mathrm{exc}}_{m}(\mathbf{r}'', \mathbf{R})\rangle_{\ell,m},\end{aligned} \tag{1.5a}$$

$$\langle \mathbf{E}^{\mathrm{exc}}_{m}(\mathbf{r}'', \mathbf{R})\rangle_{\ell,m} = \prod_{\substack{n=1\\n\neq\ell,m}}^{N} \int \mathrm{d}^3 r_n \rho(\mathbf{R}|\mathbf{r}_\ell, \mathbf{r}_m) \mathbf{E}^{\mathrm{exc}}_{n}(\mathbf{r}''), \tag{1.5b}$$

where Eq. (1.5b) represents the configurational average of the excitation field $\mathbf{E}^{\mathrm{exc}}_m(\mathbf{r}'')$ acting on the m-th particle, considering the conditional probability density $\rho(\mathbf{R}|\mathbf{r}_\ell, \mathbf{r}_m)$ that the system is in a configuration $\mathbf{R}$ with the positions $\mathbf{r}_\ell$ and $\mathbf{r}_m$ fixed in space [26, 27]. By continuing this process for all N spheres in the ensemble, a set of N equations known as the Foldy-Lax hierarchy [25] can be constructed. Solving this hierarchy leads to the solution of the averaged electric field at a point $\mathbf{r}$, taking multiple scattering effects into account. The hierarchical equations can be truncated at different orders, reducing them to invertible systems of equations by applying approximations to how multiple scattering excites the elements of the ensemble.

To truncate the Foldy-Lax hierarchy and determine the total averaged electric field, the CSM imposes two approximations at different orders. First, the Single Scattering Approximation (SSA) is considered, reducing the Foldy-Lax hierarchy to a single equation to be solved by neglecting multiple scattering effects [26]. Specifically, in the SSA, $\langle \mathbf{E}^{\mathrm{exc}}_{\ell}(\mathbf{r}'', \mathbf{R})\rangle_{\ell}$ is forced to equal the incident electric field $\mathbf{E}^{\mathrm{inc}}(\mathbf{r})$ in Eq. (1.4b) [25, 27]. The second approximation in the CSM is the Quasi-Crystalline Approximation (QSA), where the Foldy-Lax hierarchy is rewritten as a system of two equations by assuming that the configurational average for cases with one and two fixed particles is approximately equal, meaning that Eqs. (1.4b) and (1.5) are set equal [25, 26]. While the SSA is suitable for dilute disordered arrays, as it disregards multiple scattering, the QSA adapts to denser ensembles —up to a coverage fraction of 60% [25]— because, under this condition, the average with two fixed particles does not significantly differ from the average with one fixed particle due to the presence of the remaining particles [26].

Once the approximation orders have been determined to truncate the Foldy-Lax hierarchy, the CSM introduces a series of assumptions about the particle ensemble to solve the resulting system of equations for each approximation. In particular, the CSM considers that the particle ensemble consists of $N \gg 1$ identical spherical particles of radius a —embedded in a dielectric medium known as the matrix— whose centers $\mathbf{r}_\ell$ are within the integration volume V, which consists of an infinite film of thickness d centered at $z = 0$, with a uniform probability density $\rho(\mathbf{r}_\ell) = 1/V$, and assumes that the volume V_ℓ of each sphere is negligible compared to V [26]; a schematic of the system is shown in Fig. 1.1. Under these assumptions, the electric field induced by the ensemble particles in the SSA can be computed by substituting in Eq. (1.4) the dyadic Green function $\mathbb{G}$ with its plane wave representation and performing the configurational average, along with the limit $d \to 0$ to consider the case of a two-dimensional array [26]. This procedure

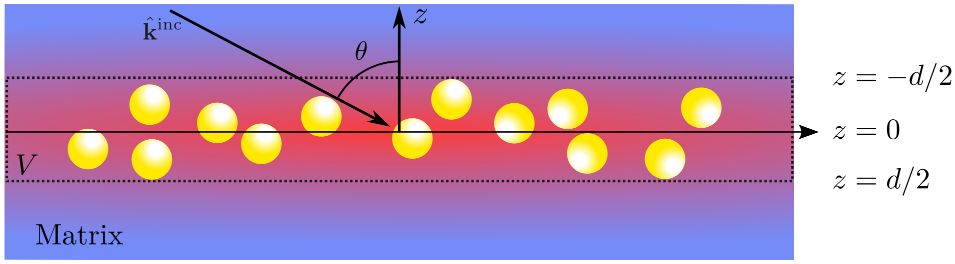

Fig. 1.1: Diagram of the particle ensemble considered in the CSM, consisting of a collection of N identical spherical particles of radius a whose centers are randomly located within the integration volume V: an infinite film centered at the plane $z = 0$ and of thickness d, which tends to zero to reproduce the case of a two-dimensional array. The system, embedded in a dielectric matrix (blue background), is illuminated by an incident monochromatic plane wave with traveling in the $\hat{\mathbf{k}}^{\rm inc}$ direction into the film at an angle θ. the red region represents the induced electric field due to the spherical particles.

results in

$$\langle \mathbf{E}^{\rm ind}_{\rm SSA}(\mathbf{r})\rangle = \begin{cases} -\alpha S_j(\pi - 2\theta)\exp(i\mathbf{k}^{\rm r}_{\rm coh}\cdot\mathbf{r})\mathbf{E}^{\rm inc}(\mathbf{r}), & z > 0, \\ -\alpha S(0)\exp\Big(i\mathbf{k}^{\rm t}_{\rm coh}\cdot\mathbf{r}\Big)\mathbf{E}^{\rm inc}(\mathbf{r}), & z < 0, \end{cases} \quad \text{with} \quad \alpha = \frac{2\Theta}{x^2\cos\theta}, \tag{1.6}$$

where θ is the incidence angle at which the incident plane wave illuminates the two-dimensional array, $S_j(\vartheta)$ are the elements of the Mie scattering matrix [29] —which depend on the refractive index of the matrix, the sphere, and its size— evaluated at ϑ, with $j = 1$ for s-polarization and $j = 2$ for p-polarization. The term Θ corresponds to the coverage fraction of the array, and $x = k^{\rm ind}a$ is the size parameter with $k^{\rm ind}$ being the magnitude of the incident field wave vector propagating in the matrix [26, 27]. The elements $S_j(\vartheta)$ are evaluated at $\vartheta = \pi - 2\theta$ and $\vartheta = 0$ as they correspond to the coherent scattering directions given by the law of reflection and Snell's law, respectively. From Eq. (1.6), it is shown that under the SSA, the two-dimensional array of spherical particles scatters light primarily in coherent directions, as the diffuse components interfere destructively [26]. Consequently, reflection $r^{(j)}_{\rm SSA}$ and transmission $t^{(j)}_{\rm SSA}$ amplitude coefficients can be defined for the two-dimensional array similarly to the Fresnel coefficients, given by

$$r^{(j)}_{\rm SSA}(\theta) = -\alpha S_j(\pi - 2\theta), \qquad \text{and} \qquad t^{(j)}_{\rm SSA}(\theta) = 1 - \alpha S(0). \tag{1.7}$$

In the case of the QSA, the system of equations to be solved consists of Eqs. (1.4) and (1.5), considering that Eqs. (1.4b) and (1.5b) are equal. To solve this system of equations, the ensemble is assumed to have the same characteristics as in the SSA case and additionally that the surroundings of each sphere are identical on average [27]. Therefore, the system of equations can be solved by proposing the *ansatz*

$$\langle \mathbf{E}^{exc}_\ell(\mathbf{r})\rangle_\ell = \mathbf{E}^{\rm exc}_{\rm r}(\mathbf{r})\ \exp(i\mathbf{k}^{\rm r}_{\rm coh}\cdot\mathbf{r}) + \mathbf{E}^{\rm exc}_{\rm t}(\mathbf{r})\ \exp\Big(i\mathbf{k}^{\rm t}_{\rm coh}\cdot\mathbf{r}\Big) \tag{1.8}$$

that is, in the CSM, it is assumed that $\langle \mathbf{E}^{exc}_\ell(\mathbf{r})\rangle_\ell$ consists of a contribution from an electric field propagating in the coherent reflection direction, denoted as $\mathbf{E}^{\rm exc}_{\rm r}$, and another in the transmission direction, denoted by $\mathbf{E}^{\rm exc}_{\rm t}$ [27]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each will have a reflected and transmitted component. The sum of these contributions determines the induced electric field under the QSA. Solving the system of equations and substituting in Eq. (1.9), the amplitude coefficients of reflection $r_{\mathrm{CSM}}^{(j)}$ and transmission $t_{\mathrm{CSM}}^{(j)}$ can be defined, considering multiple scattering [26, 27]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each of them will have a reflected and a transmitted component. The sum of these contributions determines the induced electric field under the QSA, whose expression is

$$\langle \mathbf{E}_{\mathrm{QSA}}^{\mathrm{ind}}(\mathbf{r})\rangle = \begin{cases} -\alpha\Big(S_j(0)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(\pi-2\theta)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\exp(i\mathbf{k}_{\mathrm{coh}}^{\mathrm{r}}\cdot\mathbf{r})\hat{\mathbf{E}}^{\mathrm{inc}}, & z>0, \\ -\alpha\Big(S_j(\pi-2\theta)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(0)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\exp\Big(i\mathbf{k}_{\mathrm{coh}}^{\mathrm{t}}\cdot\mathbf{r}\Big)\hat{\mathbf{E}}^{\mathrm{inc}}, & z<0, \end{cases} \tag{1.9}$$

where $\hat{\mathbf{E}}^{\mathrm{inc}}$ indicates the polarization of the incident electric field. Since in the QSA the expressions of Eq. (1.5) are equal for the electric field exciting a particle, then the contributions of the induced field in the coherent direction satisfy [27]

$$\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r}) = -\frac{\alpha}{2}\Big(S_j(0)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(\pi-2\theta)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\hat{\mathbf{E}}^{\mathrm{inc}}, \tag{1.10a}$$

$$\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r}) = \mathbf{E}^{\mathrm{inc}}(\mathbf{r}) - \frac{\alpha}{2}\Big(S_j(\pi-2\theta)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}\| + S(0)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\hat{\mathbf{E}}^{\mathrm{inc}}. \tag{1.10b}$$

Solving the system of equations for $\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})$ and $\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})$ shown in Eq. (1.10), and substituting them in Eq. (1.9), expressions for the reflection $r_{\mathrm{CSM}}^{(j)}$ and transmission $t_{\mathrm{CSM}}^{(j)}$ amplitude coefficients are defined, considering multiple scattering. These expressions are [26, 27]

$$r_{\mathrm{CSM}}^{(j)}(\theta) = \frac{-\alpha S_j(\pi-2\theta)}{1+\alpha S(0)+\frac{1}{4}\alpha^2\Big[S^2(0)-S_j^2(\pi-2\theta)\Big]}, \tag{1.11a}$$

$$t_{\mathrm{CSM}}^{(j)}(\theta) = \frac{1-\frac{1}{4}\alpha^2\Big[S^2(0)-S_j^2(\pi-2\theta)\Big]}{1+\alpha S(0)+\frac{1}{4}\alpha^2\Big[S^2(0)-S_j^2(\pi-2\theta)\Big]}, \tag{1.11b}$$

where $j=1$ for s-polarization and $j=2$ for p-polarization. The assumption that spherical particles are identical, imposed in both the SSA and QSA, can be relaxed if the radius a of the particles follows a size distribution density $\rho(a)$ and the field exciting each sphere is identical for all, except for a phase difference of $\pm 2ik^{\mathrm{inc}}a$ caused by size variation [20]. In this case, the average over the particle radius a is computed in Eq. (1.10), incorporating the phase difference by multiplying the right-hand side of Eq. (1.10) by $\exp\big(\pm 2ik_z^{\mathrm{inc}}a\big)$ —with $k_z^{\mathrm{ind}} = k^{\mathrm{ind}}\cos\theta$—, where the positive sign corresponds to the reflected contribution and the negative to the transmitted one. Following the analogous procedure described earlier, the reflection and transmission amplitude coefficients considering size polydispersity in the two-dimensional array are given by [20].

$$r_{\mathrm{CSM}}^{(j)}(\theta) = \frac{-\beta_+^{(j)}(\theta)}{1+\beta_F+\frac{1}{4}\Big[\beta_F^2-\beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\Big]}, \tag{1.12a}$$

$$t_{\mathrm{CSM}}^{(j)}(\theta) = \frac{1 - \frac{1}{4}\left[\beta_F^2 - \beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\right]}{1 + \beta_F + \frac{1}{4}\left[\beta_F^2 - \beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\right]}, \tag{1.12b}$$

where

$$\beta_F(\theta) = \eta \int_0^\infty \rho(a) S(0)\,\mathrm{d}a \quad \text{and} \quad \beta_\pm^{(j)}(\theta) = \eta \int_0^\infty \rho(a) S_j(\pi - 2\theta) \exp\left(\pm 2ik_z^{\mathrm{inc}} a\right) \mathrm{d}a\,, \tag{1.13}$$

with $\eta = 2\Theta/(x_0^2 \cos\theta)$ and where $x_0 = k^{\mathrm{ind}} a_0$ is the size parameter of a sphere with radius a_0, which is the most probable value according to the size distribution density $\rho(a)$.

The expressions in Eq. (1.12) allow describing the response of a disordered metasurface of spherical NPs under conditions close to experimental ones by introducing the presence of a substrate. To achieve this, multiple reflections between the substrate-matrix interface and the matrix-metasurface interface are considered, assuming that the latter responds according to Eq. (1.12) when separated from the substrate by a distance $\Delta \to 0$ [20]. In particular, to compare the theoretical response with reflectance measurements in an internal incidence scheme, the amplitude reflection coefficient of the entire system (metasurface with substrate) is given by [20]

$$r_{\mathrm{CSM}}^{(j)}(\theta_i) = \frac{r_{\mathrm{sm}}^{(j)}(\theta_i) + r_{\mathrm{CSM}}^{(j)}(\theta_t)}{1 + r_{\mathrm{sm}}^{(j)}(\theta_i) r_{\mathrm{CSM}}^{(j)}(\theta_t)}, \tag{1.14}$$

where r_{sm} is the Fresnel reflection coefficient between the substrate and the matrix for s (p) polarization if $j = 1$ ($j = 2$), and where θ_i is the angle of incidence of the plane wave illuminating the interface between the substrate and the matrix, and θ_t is the transmission angle upon crossing this interface and thus the angle at which it illuminates the metasurface.

1.2 Thin Film Theory

The Thin Film Theory (TFT) is theoretical framework used to describe the optical properties of thin films and small particles deposited on a substrate. The TFT is a MST that returns an approximated solution to Maxwells equation similarly to Fresnel's equations where two semi-infinite media enclose a region of the space, herby identified as a thin film, with a previously unspecified composition. To describe the optical properties of the thin film, the excess field formalism is employed thus introducing a polarization and magnetization fields of the thin the film, which can model an ensemble of metallic nano-particles through constitutive equations. This approach modifies Maxwell's equation at the interfaces of the system, thus determining explicitly the discontinuities of the electromagnetic field at the boundaries of the thin film.

To study the reflection and transmission of an electromagnetic plane wave illuminating an arbitrary thin film of thickness d centered at $z = 0$ in between two media characterized by their bulk dielectric functions $\varepsilon^{\gtrless} = (n^{\gtrless})^2$, it is proposed that the total electric field $\mathbf{E}(\mathbf{r})$ is decomposed in three contributions: inside the thin film and in the media above and below it denoted by the symbols (>) and (<), respectively. Such decomposition is written as follows

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^{>}(\mathbf{r})\Theta(z) + \mathbf{E}^{<}(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{excess}}(\mathbf{r}), \tag{1.15}$$

where the harmonic dependency in the angular frequency ω is implicitly included in the employed notation and $\Theta(z)$ is the Heaviside step function; the electric fields $\mathbf{E}^{\gtrless}$ are considered to be radiation fields. The term $\mathbf{E}^{\text{excess}}(\mathbf{r})$ is the excess electric field and its contribution is non-zero in the neighborhood of the thin film since $\mathbf{E}(\mathbf{r}) \to \mathbf{E}^{\gtrless}(\mathbf{r})$ when $z \to \pm\infty$. Thus, the whole effect of the thin film in the electromagnetic properties of the system can be described by considering the total excess electric field in the film $\mathbf{E}^{\text{film}}(\mathbf{r}, z = 0)$ in the limit when $d \to 0$, thus, yielding the expression

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^{>}(\mathbf{r})\Theta(z) + \mathbf{E}^{<}(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{film}}(\mathbf{r}_\parallel)\delta(z), \tag{1.16}$$

with $\delta(z)$ the Dirac delta function and

$$\mathbf{E}^{\text{film}}(\mathbf{r}_\parallel) = \int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r})\, \mathrm{d}z = \left(\int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) \exp(ik_z z)\, \mathrm{d}z \right)\Bigg|_{k_z=0}, \tag{1.17}$$

where $\mathbf{r}_\parallel$ is a vector evaluated at the plane $z = 0$, and where it is explicitly shown that $\mathbf{E}^{\text{film}}(\mathbf{r}_\parallel)$ equals the Fourier transform of the excess electric field in the z variable evaluated at $k_z = 0$. The same decomposition employed in Eqs. (1.15)–(1.17) for $\mathbf{E}(\mathbf{r})$ applies to the magnetic field $\mathbf{B}(\mathbf{r})$, as well as any other scalar or vector field relevant to the system, for example, the electric displacement field $\mathbf{D}(\mathbf{r})$, the $\mathbf{H}(\mathbf{r})$ field, and the external charge $\rho_{\text{ext}}(\mathbf{r})$ and current $\mathbf{J}_{\text{ext}}(\mathbf{r})$ densities. By substituting such definitions of the fields into Maxwell's equations, the following expressions for the excess quantities are obtained

$$\nabla \cdot \mathbf{D}^{\text{excess}}(\mathbf{r}) + \left[D_\perp^{>}(\mathbf{r}_\parallel) - D_\perp^{<}(\mathbf{r}_\parallel) \right] \delta(z) = \rho_{\text{ext}}^{\text{excess}}(\mathbf{r}), \tag{1.18a}$$

$$\nabla \cdot \mathbf{B}^{\text{excess}}(\mathbf{r}) + \left[B_\perp^{>}(\mathbf{r}_\parallel) - B_\perp^{<}(\mathbf{r}_\parallel) \right] \delta(z) = 0, \tag{1.18b}$$

$$\nabla \times \mathbf{E}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_\perp \times \left[\mathbf{E}_\parallel^{>}(\mathbf{r}_\parallel) - \mathbf{E}_\parallel^{<}(\mathbf{r}_\parallel) \right] \delta(z) = +i\omega \mathbf{B}^{\text{excess}}(\mathbf{r}), \tag{1.18c}$$

$$\nabla \times \mathbf{H}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_\perp \times \left[\mathbf{H}_\parallel^{>}(\mathbf{r}_\parallel) - \mathbf{H}_\parallel^{<}(\mathbf{r}_\parallel) \right] \delta(z) = \mathbf{J}_{\text{ext}}^{\text{excess}}(\mathbf{r}) - i\omega \mathbf{D}^{\text{excess}}(\mathbf{r}), \tag{1.18d}$$

where it was considered that the fields in the semi-infinite media are solution to Maxwell's equations and that the step and delta functions are related as $\mathrm{d}\Theta(\pm z)/\mathrm{d}z = \pm\delta(z)$. The subscript ($\parallel$) correspond to the parallel components of the fields relative to the thin film, ($\perp$) to their normal components to the film; $\hat{\mathbf{e}}_\perp$ is the unitary vector in this direction. The terms in between brackets in Eqs. (1.18) are the boundary conditions of the electromagnetic fields evaluated at the thin film and they are modified from the known case of a sharp interface —which yield Fresnel's equations— by the excess quantities.

The boundary conditions of the electromagnetic field in the TFT formalism can be determined by calculating the Fourier transform of Eqs. (1.18) in the z variable and then considering the particular case of $k_z = 0$ so that all excess quantities are interchanged for their total contribution according to Eq. (1.17). Additionally, total excess polarization $\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel)$ and magnetization $\mathbf{M}^{\text{film}}(\mathbf{r}_\parallel)$ are defined in terms of the total excess electromagnetic fields, thus the boundary conditions in the absence of external sources —$\mathbf{J}_{\text{ext}}(\mathbf{r}) = \mathbf{0}$ and $\rho_{\text{ext}}(\mathbf{r}) = 0$— can be written as

$$\left[D_\perp^>(\mathbf{r}_\parallel) - D_\perp^<(\mathbf{r}_\parallel)\right] = -\nabla_\parallel \cdot \mathbf{P}_\parallel^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19a}$$

$$\left[B_\perp^>(\mathbf{r}_\parallel) - B_\perp^<(\mathbf{r}_\parallel)\right] = -\nabla_\parallel \cdot \mathbf{M}_\parallel^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19b}$$

$$\left[\mathbf{E}_\parallel^>(\mathbf{r}_\parallel) - \mathbf{E}_\parallel^<(\mathbf{r}_\parallel)\right] = -i\omega\hat{\mathbf{e}}_\perp \times \mathbf{M}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \nabla_\parallel P_\perp^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19c}$$

$$\left[\mathbf{H}_\parallel^>(\mathbf{r}_\parallel) - \mathbf{H}_\parallel^<(\mathbf{r}_\parallel)\right] = +i\omega\hat{\mathbf{e}}_\perp \times \mathbf{P}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \nabla_\parallel M_\perp^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19d}$$

where $\nabla_\parallel$ and $\nabla_\parallel\cdot$ are the bidimensional gradient and divergence operators on the thin film, respectively, and

$$\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{D}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \varepsilon_0 E_\perp^{\text{film}}(\mathbf{r}_\parallel)\hat{\mathbf{e}}_\perp \tag{1.20}$$

$$\mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \frac{1}{\mu_0}\mathbf{B}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - H_\perp^{\text{film}}(\mathbf{r}_\parallel)\hat{\mathbf{e}}_\perp \tag{1.21}$$

with ε_0 (μ_0) the electric permittivity (magnetic permeability) of the vacuum. It is noteworthy that the definition of $\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel)$ in Eq. (1.20) is not just a direct identification of the procedure from Eqs. (1.18) to Eqs. (1.19) but it is based on the non-continuity of $\mathbf{D}_\parallel(\mathbf{r})$ and $E_\perp(\mathbf{r})$ across boundaries, which give rise to an excess polarization on the film unlike $D_\perp(\mathbf{r})$ and $\mathbf{E}_\parallel(\mathbf{r})$, whose continuity do not have such an effect on the system. An analogous argument is performed on the definition of the magnetization field in Eq. (1.21).

The boundary conditions of the electromagnetic fields shown in Eq. (1.19) reproduce Fresnel's equations, where the film corresponds to a planar sharp interface between two linear homogeneous media, when $\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{0}$. To introduce a thin film with a known structure, the total excess polarization and magnetization [Eqs. (1.20) and (1.21)] can be described by constitutive equations as an expansion to different orders of the electromagnetic fields. The particular case where the thin film corresponds to a collection of identical plasmonic scatters, with a size where no magnetic response is expected, the general constitutive relations to Eqs. (1.20) and (1.21) can be express to a linear order as

$$\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \gamma\left\langle \mathbf{E}_\parallel^{\text{film}}(\mathbf{r}_\parallel)\right\rangle + \beta\left\langle D_\perp^{\text{film}}(\mathbf{r}_\parallel)\right\rangle \hat{\mathbf{e}}_\perp \qquad \text{and} \qquad \mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{0} \tag{1.22}$$

where γ and β are known as the film susceptibilities, which encodes the physical properties of the ensemble of scatteres, and where the operator $\langle\cdot\rangle$ is the average on the film given as $\langle a^{\text{film}}(\mathbf{r}_\parallel)\rangle = (1/2)[a^>(\mathbf{r}_\parallel) + a^<(\mathbf{r}_\parallel)]$, for either a scalar or vector quantity $a^{\gtrless}(\mathbf{r})$.

The TFT employs the excess field formalism to determine the optical response of a metasurface embedded in a matrix and supported on a substrate, as shown in Fig. 1.2, both non-absorbing nor magnetic ($\mu^{\gtrless} = \mu_0$), by calculating amplitude reflection r_{TFT} and transmission t_{TFT} coefficients from Eqs. (1.19) and (1.22), considering that the thin film consists in a disordered ensemble of identical plasmonic spherical NPs. For a configuration of internal illumination, the electric field below the thin film is the superposition of an incident and a reflected electromagnetic plane waves $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and $\mathbf{E}_{\text{r}}(\mathbf{r})$ traveling in the $\mathbf{k}^{\text{inc}}$ and $\mathbf{k}^{\text{r}}$, respectively, within the substrate characterized by the dielectric function $\varepsilon^< = \varepsilon_i = n_i^2$, while above the electric field is given by the plane wave $\mathbf{E}^{\text{t}}(\mathbf{r})$ traveling in the $\mathbf{k}^{\text{t}}$ direction within the matrix with $\varepsilon^> = \varepsilon_t = n_t^2$. To determine the amplitude coefficients under the TFT the cases of an s and a p polarized incident

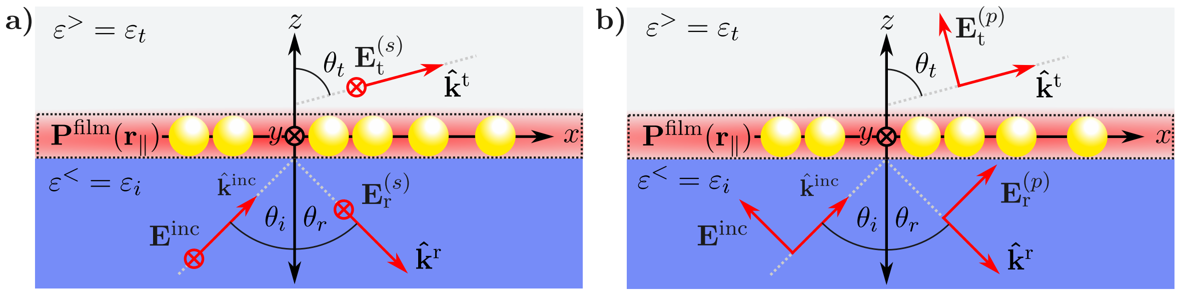

Fig. 1.2: Diagram of the reflection of transmission of **a)** an s polarized and **b)** a p polarized electromagnetic plane wave $\mathbf{E}^\mathrm{inc}$ traveling in the $\hat{\mathbf{k}}^\mathrm{inc}$ direction at an angle of incidence θ_i that illuminated a collection of identical spherical particles from a media characterized by the dielectric function $\varepsilon^< = \varepsilon_i$ (blue region) into another characterized by $\varepsilon^> = \varepsilon_t$ (grey region), where the particles are embedded. The thin film (dashed line) models the electromagnetic response of the ensemble of particles through the total excess polarization $\mathbf{P}^\mathrm{film}(\mathbf{r}_\parallel)$ (represented by the red blurred area) at $z = 0$. The reflected $\mathbf{E}_\mathrm{r}^{(j)}$ and transmitted electric field $\mathbf{E}_\mathrm{t}^{(j)}$ for a j polarization state travel at the $\hat{\mathbf{k}}_\mathrm{r}$ and $\hat{\mathbf{k}}_\mathrm{t}$ direction, respectively, that is, at the reflection θ_r and transmission angle θ_t.

electromagnetic wave are treated separately.

In general, when an electromagnetic plane wave with an angular frequency ω travels from a substrate to a thin film in the direction $\mathbf{k}^\mathrm{inc}$ at an angle θ_i, relative to the normal direction to the thin film, the reflected (transmitted) plane wave does it in the direction of $\mathbf{k}^\mathrm{r}$ ($\mathbf{k}^\mathrm{t}$) at the angle θ_r (θ_t). Thus, the wave vectors of the electromagnetic plane waves are

$$\mathbf{k}^\mathrm{inc} = n_i \frac{\omega}{c}(\sin\theta_i \hat{\mathbf{e}}_x + \cos\theta_i \hat{\mathbf{e}}_\perp), \tag{1.23a}$$

$$\mathbf{k}^\mathrm{r} = n_i \frac{\omega}{c}(\sin\theta_r \hat{\mathbf{e}}_x - \cos\theta_r \hat{\mathbf{e}}_\perp), \tag{1.23b}$$

$$\mathbf{k}^\mathrm{t} = n_t \frac{\omega}{c}(\sin\theta_t \hat{\mathbf{e}}_x + \cos\theta_t \hat{\mathbf{e}}_\perp), \tag{1.23c}$$

where $\hat{\mathbf{e}}_\perp = \hat{\mathbf{e}}_z$ and c is the speed of light. For an s polarized electromagnetic plane wave [see Fig. 1.2a)], the electric field above and below the thin film is

$$\mathbf{E}^<(\mathbf{r}) = \left[E^\mathrm{inc} \exp\left(i\mathbf{k}^\mathrm{inc}\cdot\mathbf{r}\right) + E_\mathrm{r}^{(s)} \exp(i\mathbf{k}^\mathrm{r}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_y \tag{1.24a}$$

$$\mathbf{E}^>(\mathbf{r}) = E_\mathrm{t}^{(s)} \exp\left(i\mathbf{k}^\mathrm{t}\cdot\mathbf{r}\right)\hat{\mathbf{e}}_y \tag{1.24b}$$

while for a p polarized plane wave, the electric field is given by

$$\begin{aligned}\mathbf{E}^<(\mathbf{r}) =& \left[-E^\mathrm{inc}\cos\theta_i \exp\left(i\mathbf{k}^\mathrm{inc}\cdot\mathbf{r}\right) + E_\mathrm{r}^{(p)}\cos\theta_r \exp(i\mathbf{k}^\mathrm{r}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_x + \\ &+ \left[E^\mathrm{inc}\sin\theta_i \exp\left(i\mathbf{k}^\mathrm{inc}\cdot\mathbf{r}\right) + E_\mathrm{r}^{(p)}\sin\theta_r \exp(i\mathbf{k}^\mathrm{r}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_\perp\end{aligned} \tag{1.25a}$$

$$\mathbf{E}^>(\mathbf{r}) = \left[-E_\mathrm{t}^{(p)}\cos\theta_t \exp\left(i\mathbf{k}^\mathrm{t}\cdot\mathbf{r}\right)\right]\hat{\mathbf{e}}_x + \left[-E_\mathrm{t}^{(p)}\sin\theta_t \exp\left(i\mathbf{k}^\mathrm{t}\cdot\mathbf{r}\right)\right]\hat{\mathbf{e}}_\perp \tag{1.25b}$$

as it can be seen in Fig. 1.2b). In Eqs. (1.24) and (1.25), the term $E_\xi^{(j)}$ is the magnitude of either the reflected ($\xi \to$r) or transmitted ($\xi \to$t) electric field considering the j polarization state; for either case E^inc is the magnitude of the incident electric field. Additionally, the incident $\mathbf{B}^\mathrm{inc}(\mathbf{r})$

and reflected and transmitted magnetic fields $\mathbf{B}_\xi^{(j)}(\mathbf{r})$ for both polarizations can be determined by substituing Eqs. (1.24) and (1.25) into Faraday-Lenz's Law, while the displacement field and the H field are calculated via the general relations $\mathbf{D} = \varepsilon \mathbf{E}(\mathbf{r})$ and $\mathbf{H}(\mathbf{r}) = \mathbf{B}(\mathbf{r})/\mu_0$ considered for each media. Prior calculation of the total polarization in the film $\mathbf{P}^{\mathrm{film}}(\mathbf{r}_\parallel)$ it is noteworthy to show that forcing the the proposed electric fields in Eqs. (1.24) and (1.25) to follow the boundary conditions in Eqs. (1.19) yields the continuity of the parallel component of the wave vector $\mathbf{k}_\parallel = (\omega/c) n_i \sin\theta_i \hat{\mathbf{e}}_x = (\omega/c) n_t \sin\theta_t \hat{\mathbf{e}}_x$, therefore stating that even with the presence of the thin film, the reflection and Snell's law are valid. Under this consideration, $\mathbf{P}^{\mathrm{film}}(\mathbf{r}_\parallel)$ is obtained by introducing Eqs. (1.24) and (1.25) into Eq. (1.22) yielding

$$\mathbf{P}_\parallel^{(s),\mathrm{film}}(\mathbf{r}_\parallel) = \frac{\gamma}{2}\left(E^{\mathrm{inc}} + E_{\mathrm{r}}^{(s)} + E_{\mathrm{t}}^{(s)}\right) \exp\left(\mathbf{k}_\parallel \cdot \mathbf{r}\right) \hat{\mathbf{e}}_x, \tag{1.26a}$$

$$P_\perp^{(s),\mathrm{film}}(\mathbf{r}_\parallel) = 0, \tag{1.26b}$$

for s polarization and

$$\mathbf{P}_\parallel^{(p),\mathrm{film}}(\mathbf{r}_\parallel) = \frac{\gamma}{2}\left[\left(E_{\mathrm{r}}^{(p)} - E^{\mathrm{inc}}\right)\cos\theta_i - E_{\mathrm{t}}^{(p)}\cos\theta_t\right] \exp\left(\mathbf{k}_\parallel \cdot \mathbf{r}\right) \hat{\mathbf{e}}_x, \tag{1.27a}$$

$$P_\perp^{(p),\mathrm{film}}(\mathbf{r}_\parallel) = \frac{\beta}{2}\left[\varepsilon_i\left(E^{\mathrm{inc}} + E_{\mathrm{r}}^{(p)}\right)\sin\theta_i + \varepsilon_t E_{\mathrm{t}}^{(p)}\sin\theta_t\right] \exp\left(\mathbf{k}_\parallel \cdot \mathbf{r}\right), \tag{1.27b}$$

for p polarization.

As stated above, the TFT returns expressions for reflection and transmission amplitude coefficients. These coefficients are obtained for an s polarized incident plane wave by solving the system of equations obtained by substituting Eqs. (1.23), (1.24) and (1.26) into the boundary conditions in Eq. (1.19), yielding

$$E^{\mathrm{inc}} + E_{\mathrm{r}}^{(s)} = E_{\mathrm{t}}^{(s)}, \tag{1.28a}$$

$$\left(n_i\cos\theta_i + \frac{i}{2}\frac{\omega}{c}\gamma\right)E^{\mathrm{inc}} - \left(n_i\cos\theta_i - \frac{i}{2}\frac{\omega}{c}\gamma\right)E_{\mathrm{r}}^{(s)} = \left(n_t\cos\theta_t - \frac{i}{2}\frac{\omega}{c}\gamma\right)E_{\mathrm{t}}^{(s)}. \tag{1.28b}$$

Analogously, the system of equations for a p polarized incident plane wave obtained from Eqs. (1.19), (1.23) (1.25) and (1.27) is

$$\left(\cos\theta_i + ik_\parallel\frac{\beta}{2}\varepsilon_i\sin\theta_i\right)E^{\mathrm{inc}} - \left(\cos\theta_i - ik_\parallel\frac{\beta}{2}\varepsilon_i\sin\theta_i\right)E_{\mathrm{r}}^{(p)} = \left(\cos\theta_t - ik_\parallel\frac{\beta}{2}\varepsilon_t\sin\theta_t\right)E_{\mathrm{t}}^{(p)}, \tag{1.29a}$$

$$\left(n_i + ik_\parallel\frac{\gamma}{2}\cos\theta_i\right)E^{\mathrm{inc}} + \left(n_i - ik_\parallel\frac{\gamma}{2}\cos\theta_i\right)E_{\mathrm{r}}^{(p)} = \left(n_t - ik_\parallel\frac{\gamma}{2}\cos\theta_t\right)E_{\mathrm{t}}^{(p)}, \tag{1.29b}$$

with $k_\parallel = (\omega/c) n_i \sin\theta_i = (\omega/c) n_t \sin\theta_t$. Thus, the reflection and transmission amplitude coefficients given by the solution to Eqs. (1.28) for $\xi_{\mathrm{TFT}}^{(s)} = E_\xi^{(s)}/E^{\mathrm{inc}}$ are

$$r_{\mathrm{TFT}}^{(s)}(\theta_i) = \frac{n_i\cos\theta_i - n_t\cos\theta_t + i(\omega/c)\gamma}{n_i\cos\theta_i + n_t\cos\theta_t - i(\omega/c)\gamma}, \tag{1.30a}$$

$$t_{\mathrm{TFT}}^{(s)}(\theta_i) = \frac{2n_i\cos\theta_i}{n_i\cos\theta_i + n_t\cos\theta_t - i(\omega/c)\gamma}, \tag{1.30b}$$

while for $\xi^{(p)}_{\rm TFT} = E^{(p)}_{\xi}/E^{\rm inc}$, obtained by solving Eqs. (1.29), the amplitude coefficients are

$$r^{(p)}_{\rm TFT}(\theta_i) = \frac{\kappa_-(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t + i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2\theta_i}{\kappa_+(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2\theta_i}, \tag{1.31a}$$

$$t^{(p)}_{\rm TFT}(\theta_i) = \frac{n_i\cos\theta_i\left(1-(\omega/2c)^2\varepsilon_i\gamma\beta\sin^2\theta_i\right)}{\kappa_+(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2\theta_i}, \tag{1.31b}$$

$$\kappa_\pm(\omega) = \left(n_t\cos\theta_i \pm n_i\cos\theta_t\right)\left[1-\left(\frac{\omega}{2c}\right)^2\varepsilon_i\gamma\beta\sin^2\theta_i\right]. \tag{1.31c}$$

To completely describe the optical properties of a metasurface of identical plasmonic scatters suported by a substrate with Eqs. (1.30) and (1.31), the film susceptibilities γ and β defined in Eq. (1.22) must be specified. When considering spherical scatterers of radii a embedded in the transmission medium and supported on a planner interface with the incidence medium, the electromagnetic response of the ensemble can be modeled by electric point dipoles and their images if the condition $n_t a\omega/c \ll 1$ is met. Under this considerations in addition to setting the thickness of the thin film modeling the ensemble of particles as $2a$, the film susceptibilities are given by

$$\gamma = \frac{2}{3}\frac{\Theta}{V_{\rm sp}}\left(\frac{\alpha_{\rm sp}}{1+A\alpha_{\rm sp}}\right) \qquad \text{and} \qquad \beta = \frac{2}{3}\frac{\Theta}{\varepsilon_i^2 V_{\rm sp}}\left(\frac{\alpha_{\rm sp}}{1+2A\alpha_{\rm sp}}\right) \tag{1.32}$$

where Θ the surface coverage of the particles ensemble on the substrate and where $\alpha_{\rm sp}$ and A are the polarizability of the spherical particles and the image strength, respectively, defined as

$$\alpha_{\rm sp} = 3\varepsilon_t V_{\rm sp}\left(\frac{\varepsilon_{\rm sp}-\varepsilon_t}{\varepsilon_{\rm sp}+2\varepsilon_t}\right) \qquad \text{and} \qquad A = \frac{\varepsilon_i-\varepsilon_t}{\varepsilon_i+\varepsilon_t}, \tag{1.33}$$

with $\varepsilon_{\rm sp}$ the dielectric function of the spheres and $V_{\rm sp}$ their volume. It is noteworthy that $r_{\rm TFT}$ and $t_{\rm TFT}$ describe the optical properties of the ensemble when the incident plane wave travels from the media where the spherical particles are located by interchanging $\varepsilon_t \leftrightarrow \varepsilon_i$ and $n_t \leftrightarrow n_i$ in Eqs. (1.30)–(1.33).

1.3 Experimental Setup

In order to characterize and analyze the optical response of a metasurface of spherical gold nanoparticles (AuNPs) with a disordered spatial distribution supported on a glass substrate, the extinction and reflectivity spectra of the metasurface are measured. In Fig. 1.3 the setup used for the extinction measurements is shown, while in Fig. 1.4 it is presented the setup for the reflectivity measurements.

The extinction measurements, as schematized in Fig. 1.3a), are obtained by illuminating the metasurface with a white light source (LS) at normal incidence at an external configuration, that is, from air into the substrate. The light is collimated with a lens (L) before the AuNPs metasurface and then transmitted light is collected with a glass fiber connected to an spectrometer

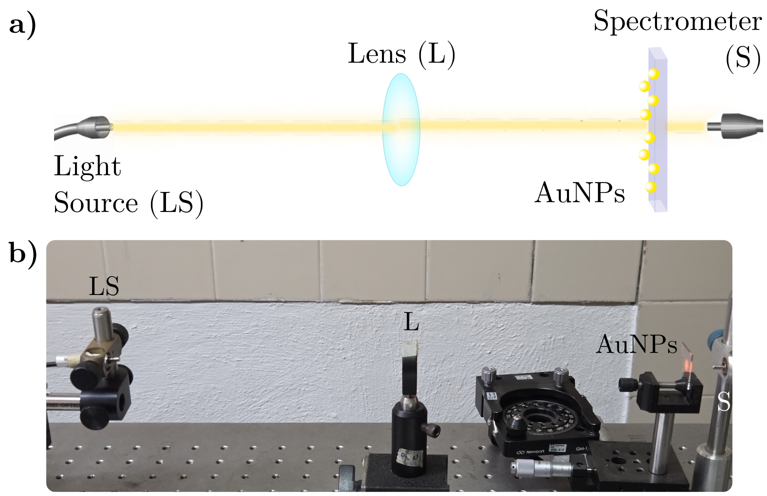

Fig. 1.3: **a)** Schematic representation and **b)** photograph of the experimental setup to perform extinction spectra measurements. The setup consists in a white light source (LS), whose light beam is collimated by a lens (L) before illuminating a metasurface of AuNPs at normal incidence. The transmitted light through the metasurface is collected into a spectrometer (S).

(S). An image of the experimental setup is presented in Fig. 1.3b) where each component of the setup is signalized. To perform the extinction measurements, it was employed a halogen lamp (Mikropack HL-2000) as the white light source and a StellarNet(R) BLUE-Wave Spectrometer with spectral range between 350 nm and 1150 nm.

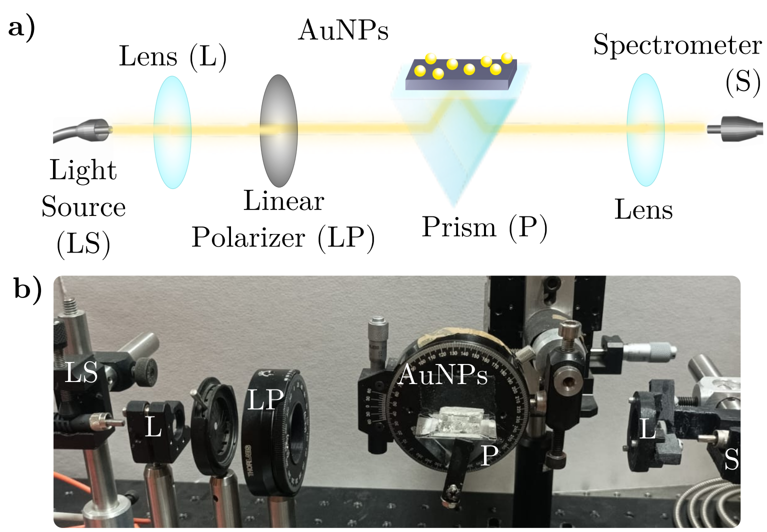

Fig. 1.4: **a)** Schematic representation and **b)** photograph of the experimental setup to perform reflectivity spectra measurements. The setup consists in a white light source (LS), whose light beam passes through a lens (L) and linear polarizer (LP) before incising a glass prism (P). The metasurface of AuNPs is coupled with an immersion oil so the AuNPs are illumited from inside the prism at the desired angle of incidence. The reflected light is collected with a lens (L) into a spectrometer (S).

The reflectivity spectra is measured in an internal illumination configuration with a white light beam from a Xenon lamp (ThorLabs SLS201L) as a light source (LS). In Fig. 1.4a) it is shown the schematic representation of the experimental setup presented in Fig. 1.4b), where it can be seen that the light beam passes through a lens (L) and a linear polarizer (LP) before incising into a planar face of a right-angle prism (P) supporting the metasurface. The reflected beam from the metasurface is then collected by a second lens (L) and directed into a spectrometer (S): Ocean Optics HR4000 Spectrometer. To perform the reflectivity measurements, the prism (BK7 glass) and the substrate of the metasurface (Corning glass) were optically coupled by index-matching their materials with a microscope immersion oil with a refractive index of $n = 1.518$.

A scheme of the optically coupled prism and the metasurface is shown in Fig. 1.5a) where the red line corresponds to the optical path of the light beam through the prism into the AuNPs, the cyan line to the immersion oil index matching the metasurface and the prism, and where the dashed gray lines represent the normal direction to the planar faces of the prism. The light beam travels into the prism at an angle θ_g, experimentally determined by a goniometer holding the prism as shown in Fig. 1.5b), which determines the angle of incidence θ_i at which the AuNPs are illuminated. The relations between θ_g and θ_i are

$$\theta_i = \frac{\pi}{4} + \sin^{-1}\left(\frac{n_{\text{Air}}}{n_{\text{Prism}}}\sin\theta_g\right) \qquad \text{and} \qquad \theta_g = \sin^{-1}\left[\frac{n_{\text{Prism}}}{n_{\text{Air}}}\sin\left(\theta_i - \frac{\pi}{4}\right)\right],$$

where n_{Air} is the refractive index of the air (typically considered equal the unity) and n_{Prism} that of the prism. Due to the index matching, the refraction between the prism and the matasurfaces's substrate are neglected.

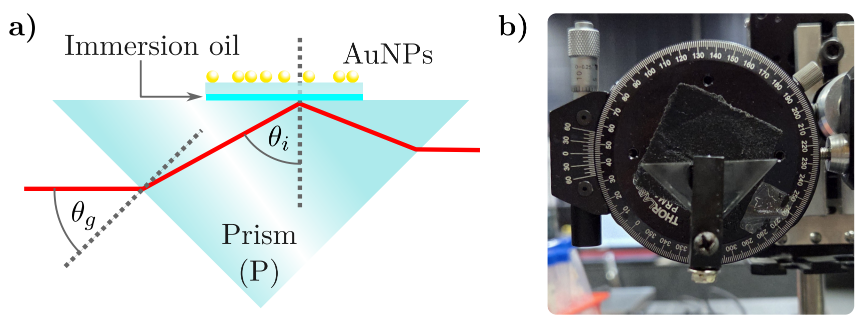

Fig. 1.5: a) Schematic representation of the right-angle prism (P) mounted in the reflectivity measurement setup described in Fig. 1.4 and **b)** photograph of the prism attached to a goniometer used to control the angle of incidence θ_i of a light beam on the metasurface of AuNPs (not shown in the photograph). A light beam (red line) entering the prism at an angle θ_g, measured with the goniometer scale, is refracted inside the prism according to Snell's law. This refraction determines θ_i at the interface between the prism and the metasurface's substrate, which is optically coupled using an immersion oil layer (cyan line). The dashed gray lines corresponds to the normal direction to any of the planar faces of the prism.

Chapter 2

Progress in non-intrusive characterization

Una iontroducción a esta sección

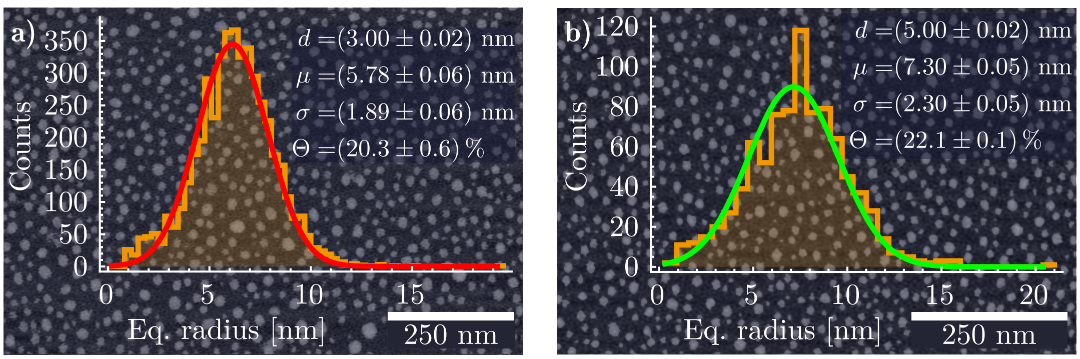

Fig. 2.1: SEM image and histogram of the equivalent radius of the AuNPs in a metasurface fabricated from a gold film with an initial thickness **a)** $d = (3.00 \pm 0.02)$ nm and **b)** $d = (5.00 \pm 0.02)$ nm, both showing a normal distribution of the equivalent radius of the AuNPs with a mean μ, standard deviation σ, and surface coverage Θ.

2.1 Substrate effects on a plasmonic metasurface

Una iontroducción a esta sección

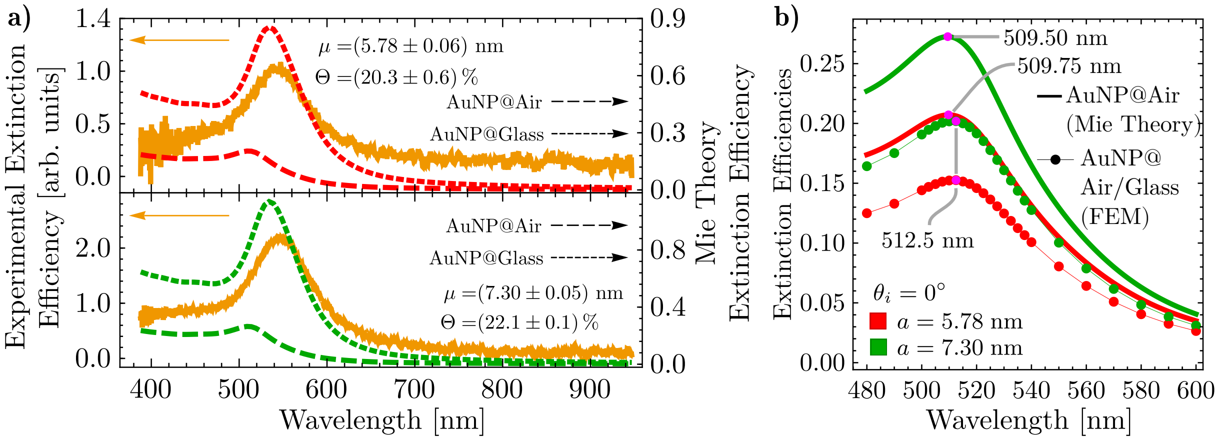

Fig. 2.2: a) Experimental extinction efficiency of the metasurfaces (continuous orange lines) when illuminated at normal incidence, and extinction efficiency of a single AuNP of radius $a = 5.78$ nm (red lines) and $a = 7.30$ nm (green lines) embedded in air (long dashed lines) and in glass (short dashed lines), as given by Mie Theory, as a function of the incident light wavelength. **b)** Extinction efficiencies as a function of the incidence wavelength, at normal incidence, of a single AuNP embedded in an infinite air matrix (continuous lines) calculated by means of Mie Theory, and in a semiinfinite air matrix supported on a semiinfinte glass substrate (markers) using FEM simulations. The red lines correspond to the calculations of a AuNP with radius $a = 5.78$ nm and the green lines to $a = 7.30$ nm. The magenta markers show the excitation of the LSPR for each case.

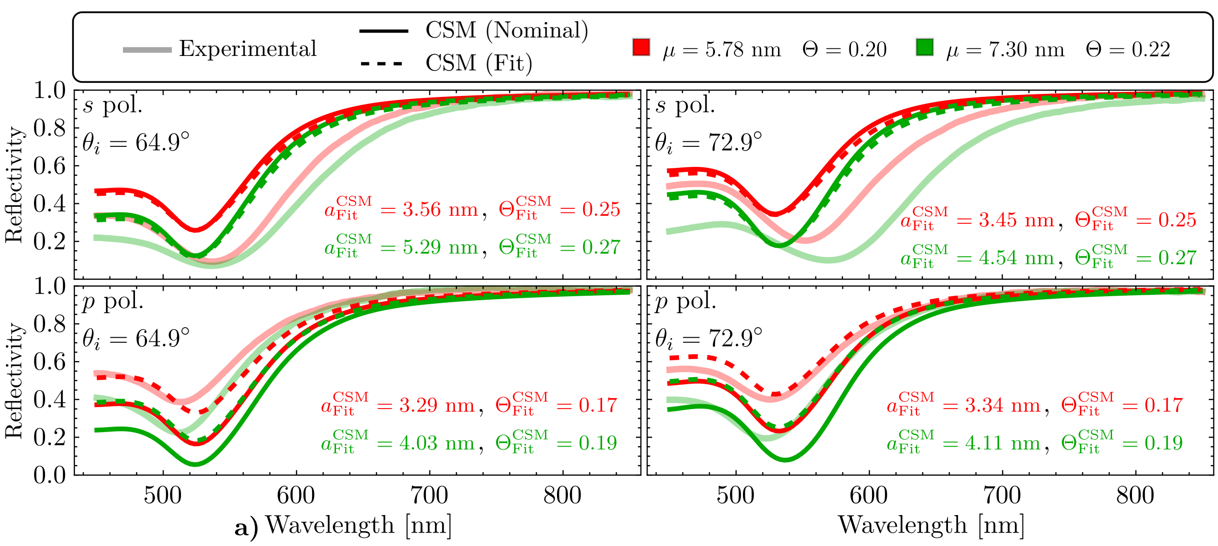

Fig. 2.3: Reflectivity at ATR configuration of a AuNPs metasurface as a function of the incident wavelength considering an angle of incidence $\theta_i = 64.9°$ (left column) and $\theta_i = 72.9°$ (right column), for s (upper row) and p (lower row) polarization state. The metasurface is embedded in a water matrix and supported onto a glass substrate with refractive indices $n_{\mathrm{m}} = 1.33$ and $n_{\mathrm{s}} = 1.5$, respectively. The experimental reflectivity (continuous translucent lines) of the metasurface is shown for the fabricated samples characterized by the mean radius μ of their size distribution and by its surface coverage Θ. The red lines correspond to the metasurface shown in Fig. 2.1**a)** while the green lines to the sample shown in Fig. 2.1**b)**. The theoretical curves (opaque continuous and dashed lines) were calculated by the CSM considering the nominal values of the metasurface, and a mean radius a^{CSM} for the size distribution and a surface coverage Θ^{CSM}, respectively; these last values were obtained by fitting the CSM to the experimental data through the descent gradient method.

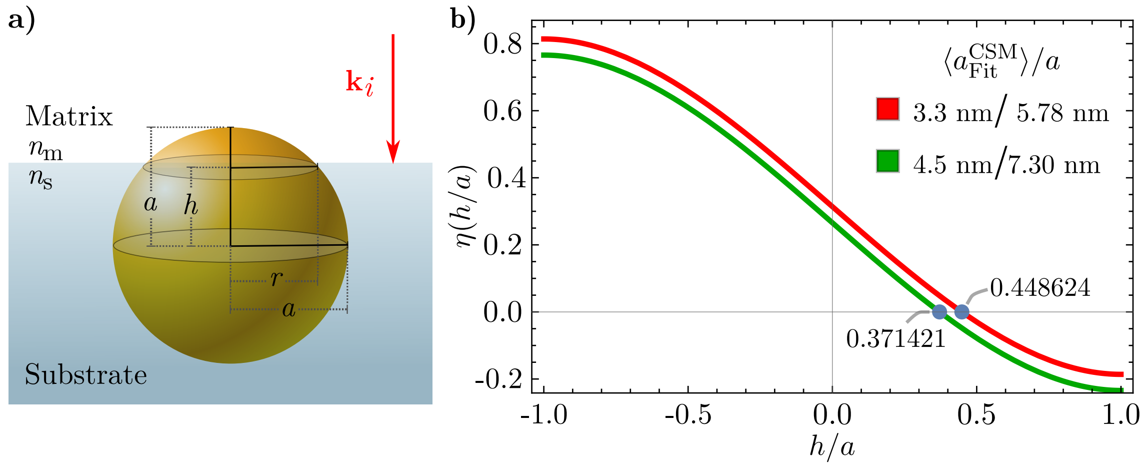

Fig. 2.4: a) Schematics of a AuNP of radius a partially embedded in a substrate, illuminated by an electromagnetic plane wave with an incident wave vector $\mathbf{k}_i$ perpendicular to the matrix-substrate interface. The distance between the center of the AuNP and the interface is defined as h while r is the aparent radius of the AuNP as seen from above. **b)** Plot of function $\eta(h/a)$ defined in Eq. (**??**) for a metasuface characterized for the most probable radius in its size distribution $a = 5.78$ nm and the average of the best fittig radius to the CSM $\langle a_\mathrm{Fit}^\mathrm{CSM}\rangle = 3.3$ nm (red lines) and for a metasurfae with $a = 7.30$ nm and $\langle a_\mathrm{Fit}^\mathrm{CSM}\rangle = 4.5$ nm (green lines). The roots of $\eta(h/a)$ are signilized with the blue markers for each case.

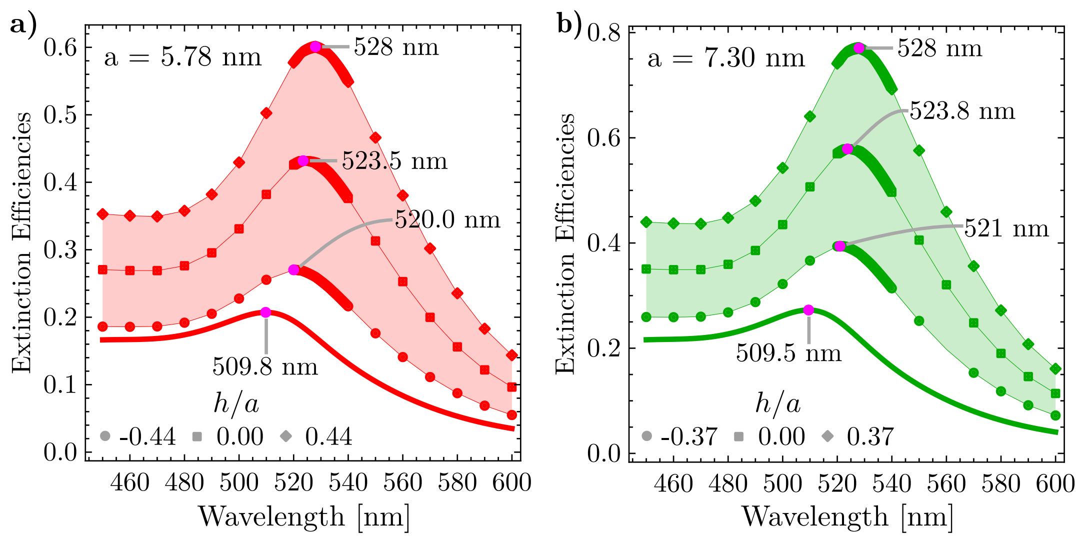

Fig. 2.5: Extinction efficiencies as function of the incident wavelength of a single AuNP, partially embedded in a glass substrate and an air matrix, of radius **a)** $a = 5.78$ (red lines) nm and **b)** $a = 7.30$ nm (green lines) when illuminated at normal incidence relative to the air-glass interface. The partial embedding of the AuNPs is characterized by the quantity h/a whose nominal value is specified for each AuNP according to Fig. **??**. For a more complete analysis, three values of partial embedding where considered: when the AuNP is mostly embedded in air (disk), in glass (diamond) and when it is equally embedded in both media (square).

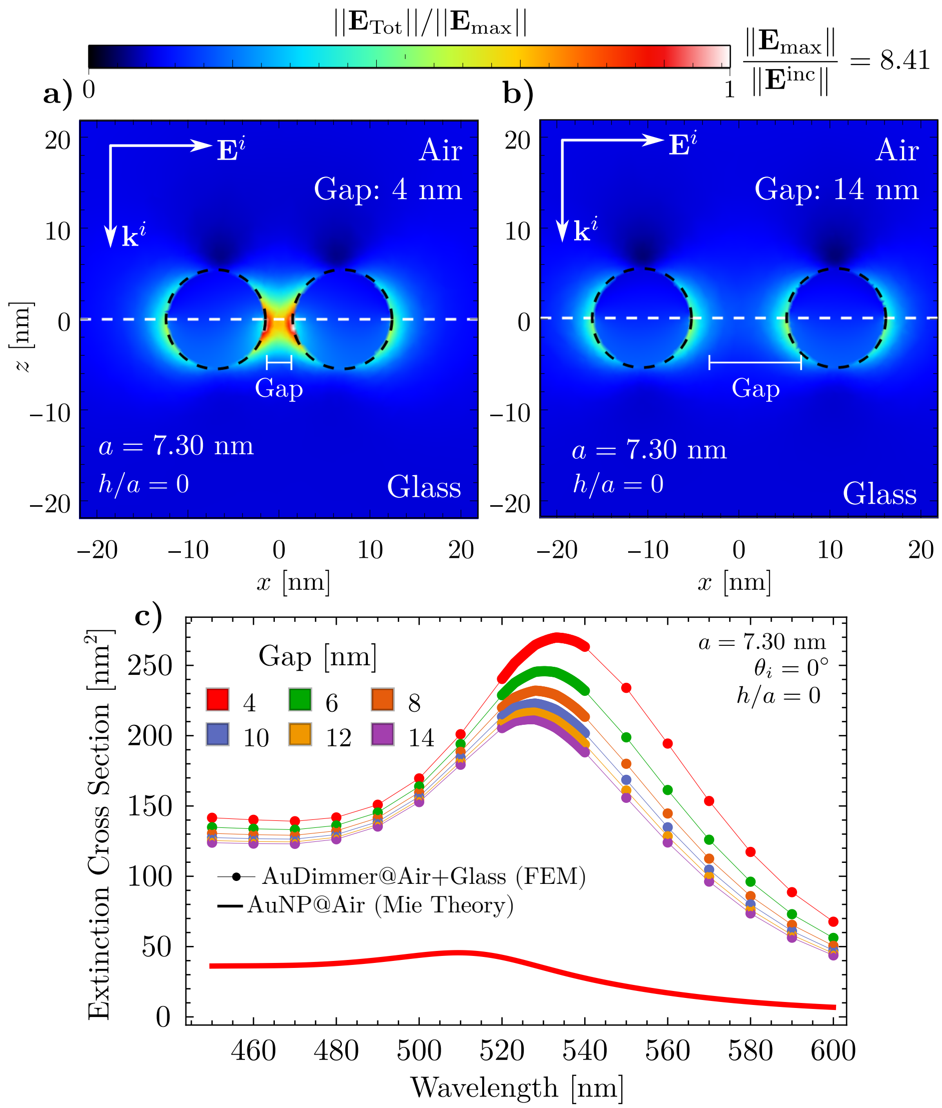

Fig. 2.6: Magnitude of the total electric field $||\mathbf{E}_{\text{Tot}}||$ of a AuNP dimmer of radius $a = 7.30$ nm separaeted by a gap between particles of **a)** 4 nm and **b)** 14 nm, illuminated by an incident electromagnetic plane wave traveling normal to a glass-air interface (white dashed line). The AuNP is partially embedded in the glass substrate and in the air matrix and its embedding degree is characterized by the quantity $h/a = 0$.**c)**xtinction cross section of a dimmer composed of two indentical AuNP of radius $a = 7.30$ nm. The extinction cross section is presented as function of the incident wavelength of an electromagnetic plane polarized along the symmetry axis of the dimmer. The AuNPs are partially embedded in an air a matrix and a glass substrate considering $h/a = 0$, and the dimmer is illuminated at normal incidence relative to the air-glass interface. The color of each curve corresponds to different values of the in between the AuNPs of the dimmer.

Chapter 3

Progress in sensing mechanisms

Una iontroducción a esta sección

3.1 Firsts Measurements

Una iontroducción a esta sección

3.2 Expected results: Topological Darkness

Una iontroducción a esta sección

Chapter 4

Work plan (proposed activities to complete the project)

Una iontroducción a esta sección

4.1 Research Progress

4.2 Research Timeline

List of Publications

During my doctoral studies, I contributed to the publication of the following peer-reviewed work:

Phase Singularity in Random Gold Metasurface for Refractive Index Sensing: Experimental Demonstration and Theoretical Analysis

J. P. Cuanalo-Fernández, N. Korneev, I. Cosme, M. B. De La Mora, **J. A. Urrutia-Anguiano**, A. Reyes-Coronado, R. Ramos-García, and S. Mansurova

J. Phys. Chem. C **128**(27): 11372–11381, 2024.

DOI: 10.1021/acs.jpcc.4c02274

The phenomenon of topological darkness has been experimentally observed and theoretically characterized in metasurfaces with an ordered spatial distributions of their building entities and exploited as a high sensitive sensing mechanism. In this work, the conditions to obtain topological darkness —near zero reflectance and phase singularity— are met in a random array of gold nanoislands (oblate ellipsoids) using theoretical and experimental methods. Reflectance and phase spectra are measured using a common-path interferometer in an attenuated total reflectance setup. Results reveal a partial state of topological darkness, with phase singularities marked by abrupt $\pm\pi$ phase jumps. Additionally, the study highlights the potential of phase and derivative measurements near the singularity for highly sensitive refractive index detection. The findings align well with theoretical predictions based on the Thin Film Theory.

Within the collaborations, I am working on the calculations and writing process of two manuscripts for submission as peer-reviewed publications:

- **Title:** Theoretical and numerical approach to substrate effects in random plasmonic metasurfaces
  Authors: Jonathan A. Urrutia-Anguiano, Isabel Y. Rojas-Martinez, Juan P. Cuanalo-Fernández, Alejandro Reyes-Coronado, Rubén Ramos-García, and Svetlana Mansurova
  Status: Review between collaborators
  Description: When analyzing the optical response of metasurfaces, discrepancies between experimental measurements and theoretical predictions are often qualitatively attributed to the influence of the substrate. In this study, we compare experimentally obtained reflectivity spectra of disordered metasurfaces composed of spherical gold nanoparticles (AuNPs) with

predictions from the Coherent Scattering Model (CSM). Deviations from the experimental results are further examined through Finite Element Method (FEM) simulations of a single AuNP. By quantitatively estimating the contribution of multiple scattering and of image multipoles induced in the substrate, our findings, supported by experimental data, FEM simulations, and CSM predictions suggest that the gold nanoparticles are partially embedded in the substrate. This partial embedding effectively alters the apparent size of the nanoparticles and increases the surface coverage of the metasurface.

- **Title:** Spectral shift in the strong couple regime: Red- and blueshift in plasmonic disordered metasurfaces
  Authors: Juan P. Cuanalo-Fernández, **Jonathan A. Urrutia-Anguiano**, Irving Gazga-Gurrión, Nikolai Korneev, Alejandro Reyes-Coronado, Rubén Ramos-García, and Svetlana Mansurova
  Status: Writting process
  Description: The optical response of metasurfaces composed of disordered gold nanoparticles can be modeled using multiple scattering theories, such as the Coherent Scattering Model and the Thin Film Theory. Under Attenuated Total Reflection conditions, the resonance of these metasurfaces exhibited a spectral shift in response to changes in the refractive index of the surrounding medium of the nanoparticles. In addition to the well-known redshift, both theoretical predictions and experimental observations have also revealed a blueshift. This blueshift is believed to be an effect of strong coupling, as it occurs only when the angle of incidence is near the critical angle. Additionally, from a theoretical framework, it is studied if the blueshift is modeled only by multiple scattering theories, or if effective medium theories can reproduce such spectral behavior.

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